

- Battery (VBAT)
- Always on 1.8V up to 750mA (1V8)
- System voltage between VBAT and VUSB (Vsys)
- Three software-controlled digital supplies (VD0, VD1, VD2)
- Two software-controlled analog supplies (VA0, VA1)

- Default
- VD0: Off
- VD1: 1.8V or higher (Buck-Boost)
- VD2: Off
- VA0: Off
- VA1: 1.5 or 3.0V depending on config
- 1V8: Always On

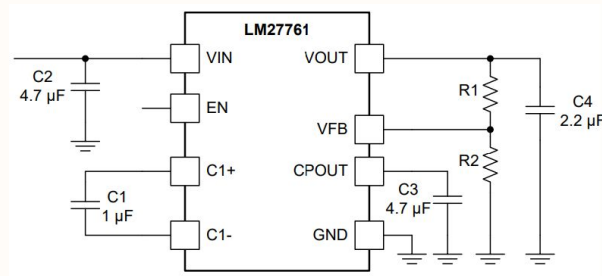
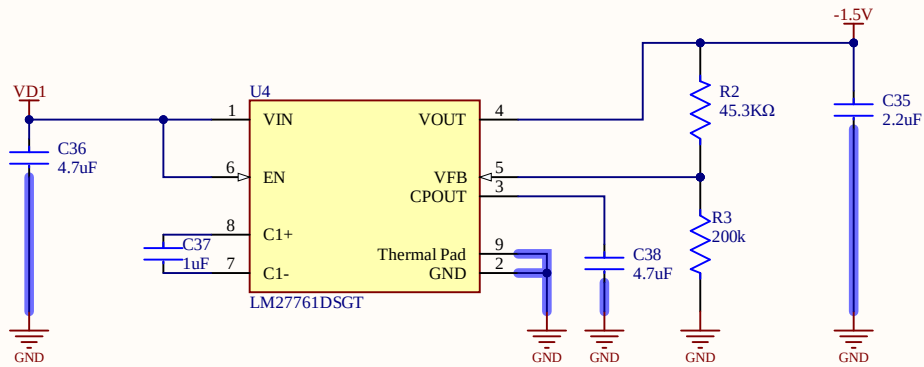
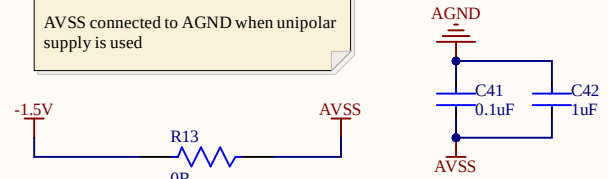


Figure 16. LM27761 Typical Application

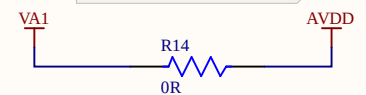
$$V_{OUT} = -1.22 \text{ V} \times (R1 + R2) / R2$$

Analog supply for the ADS can be switched on/off software controlled with the PMIC.

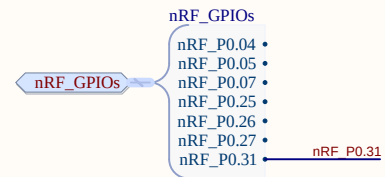
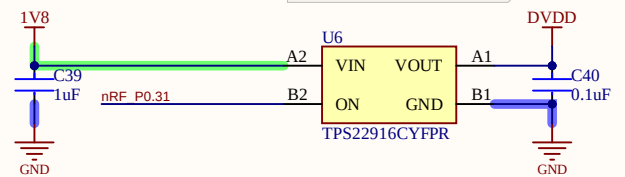
AVSS = -1.5V when bipolar supply is used
AVSS connected to AGND when unipolar supply is used

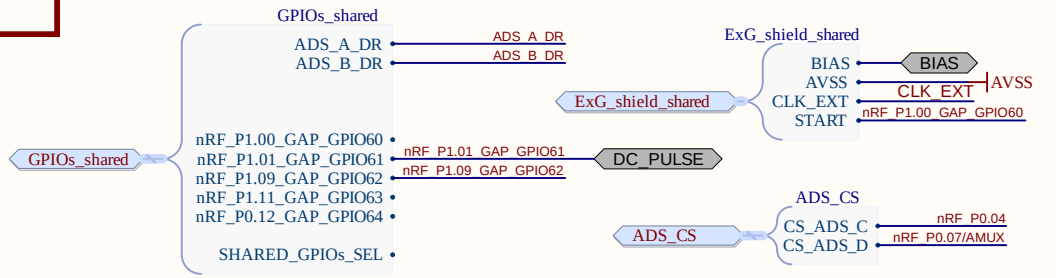
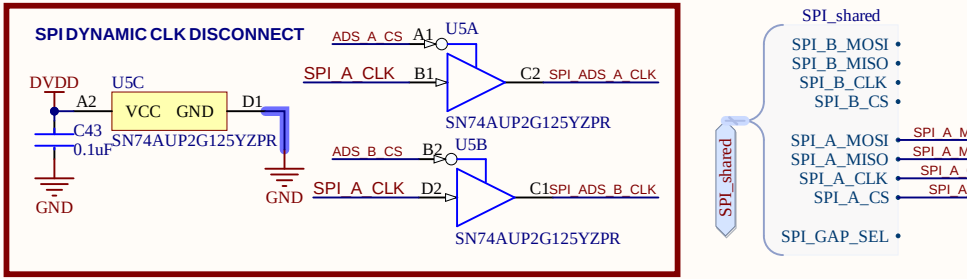
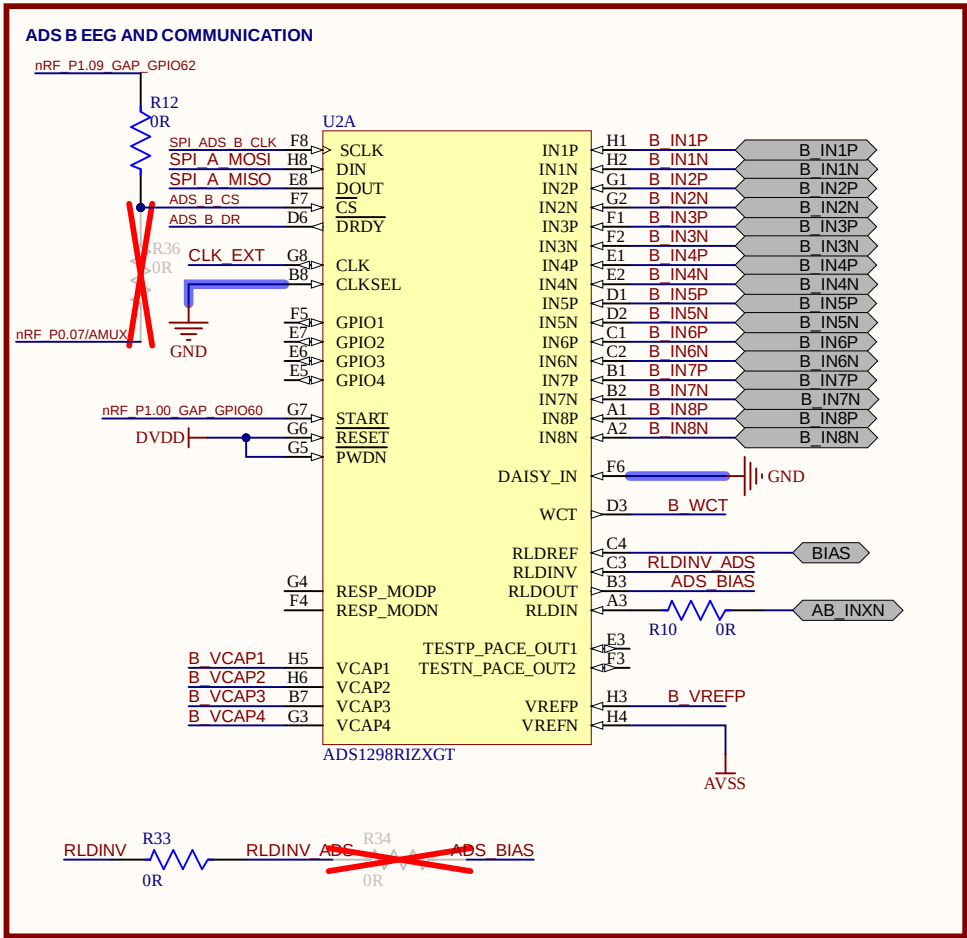
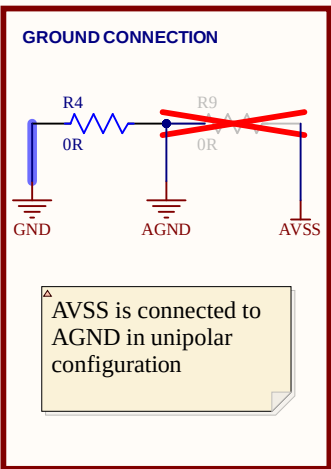
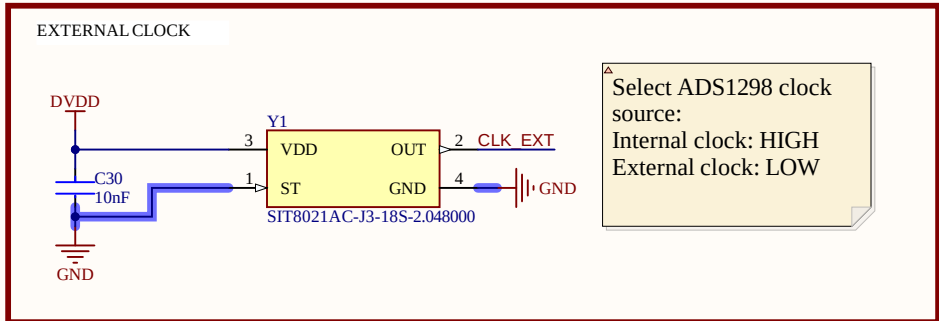
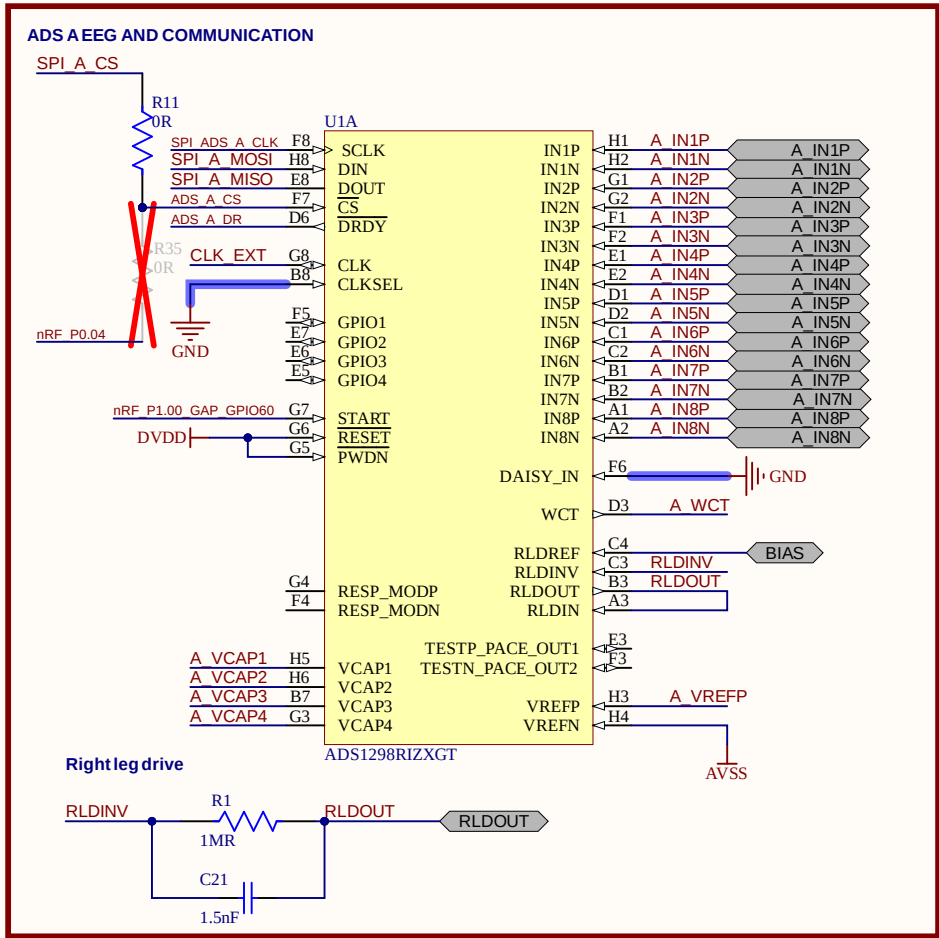
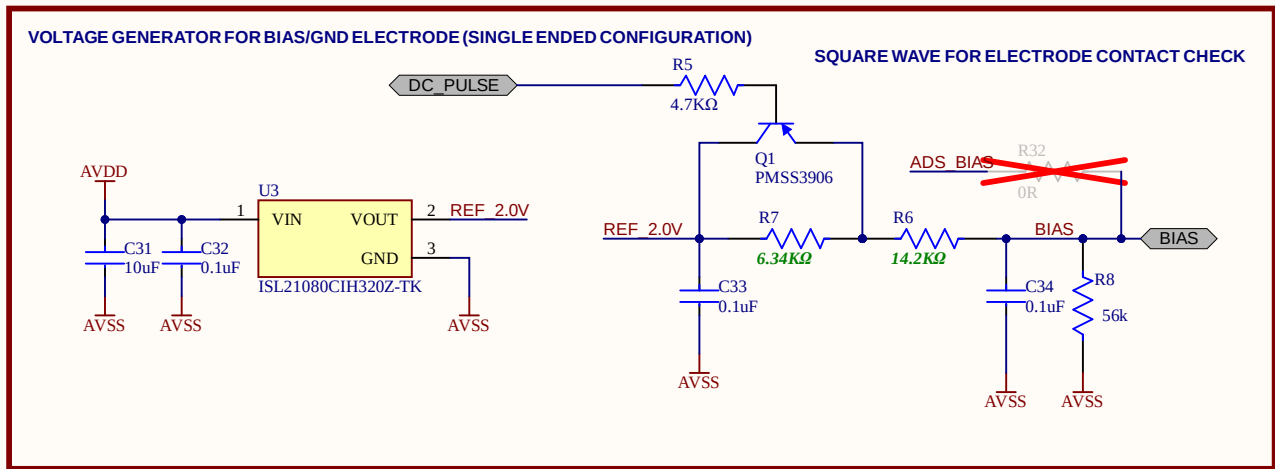
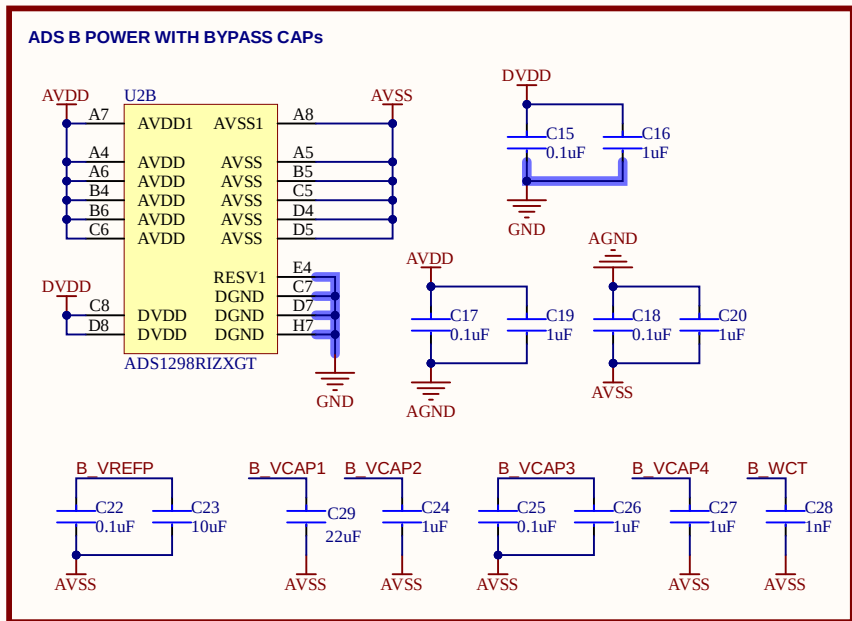
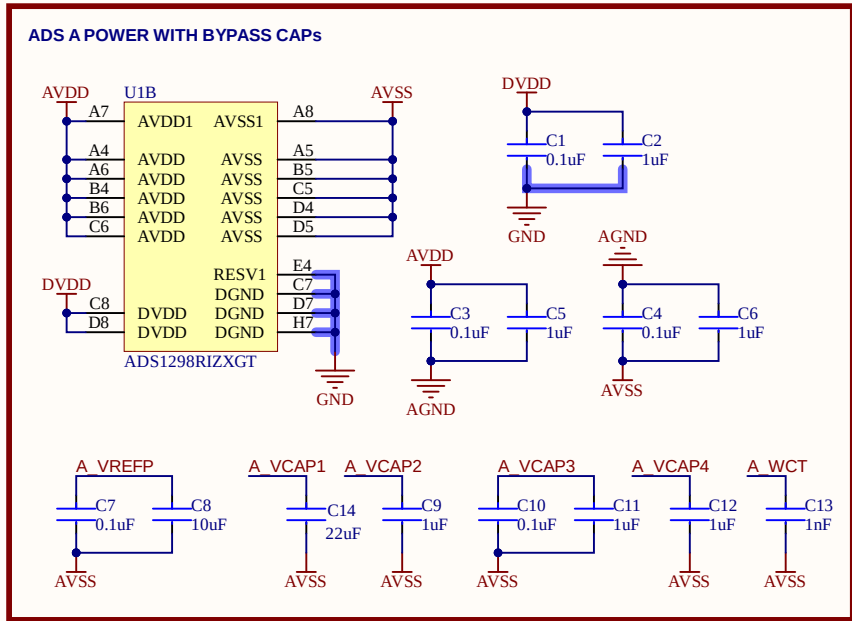


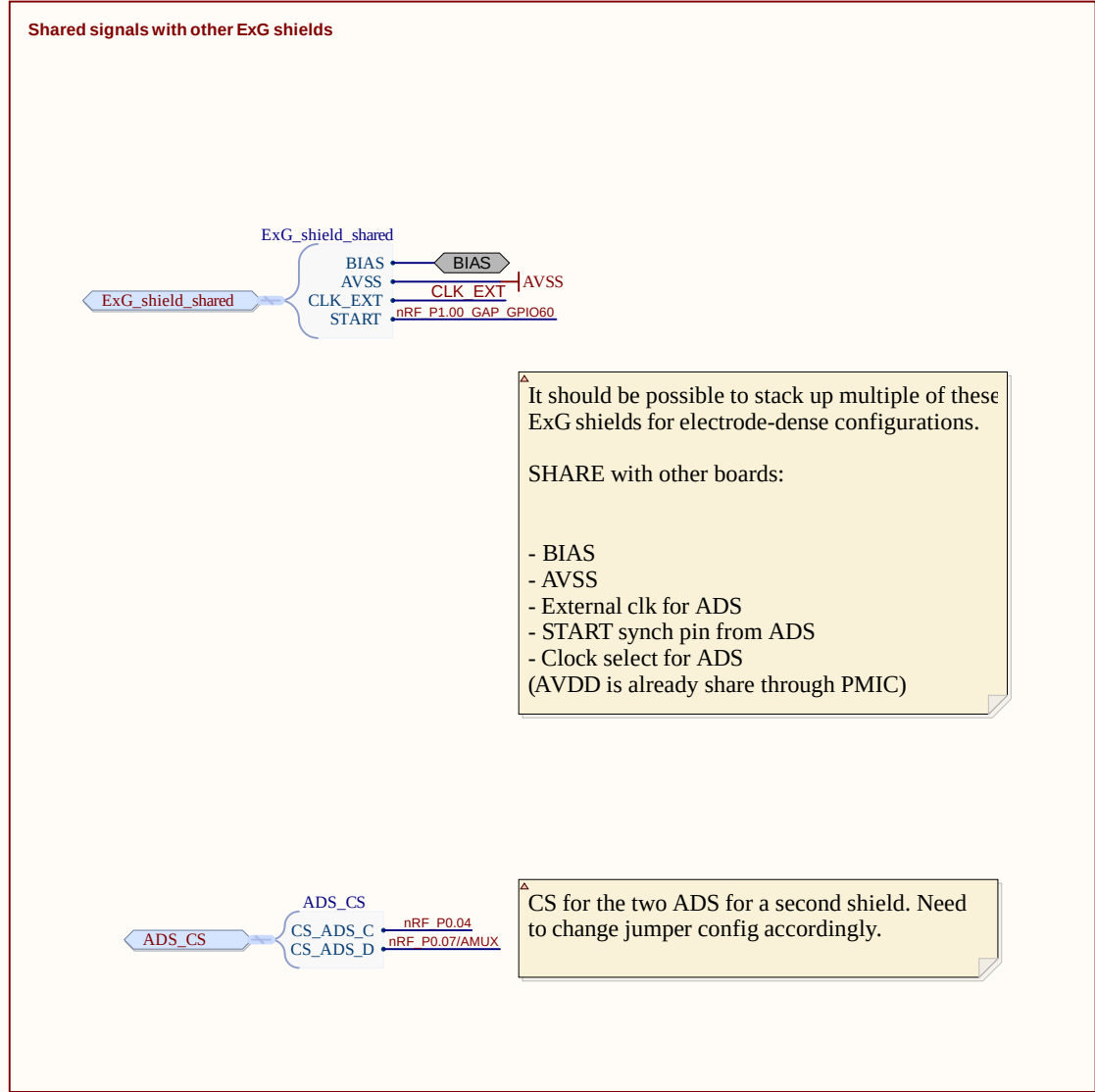
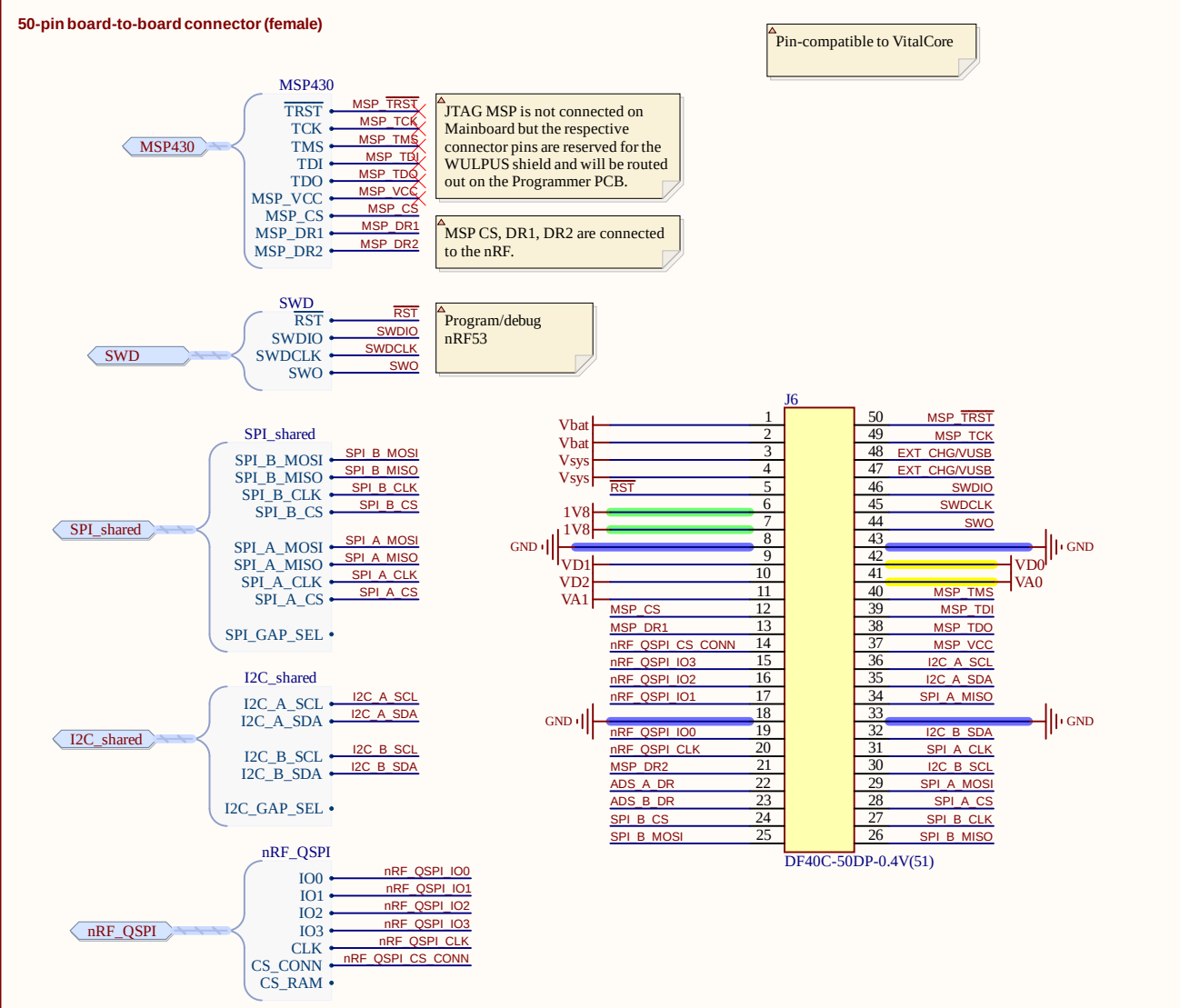
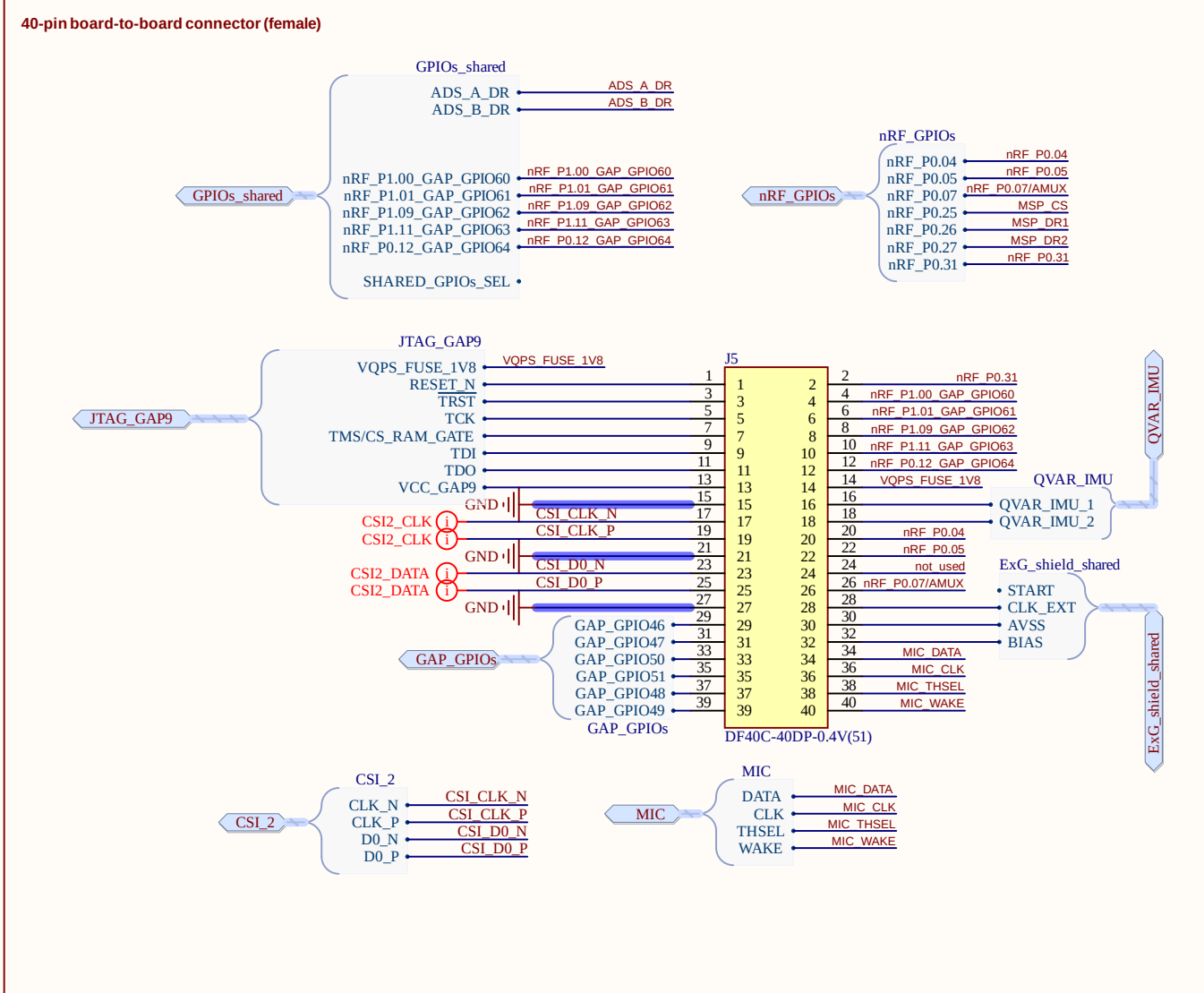
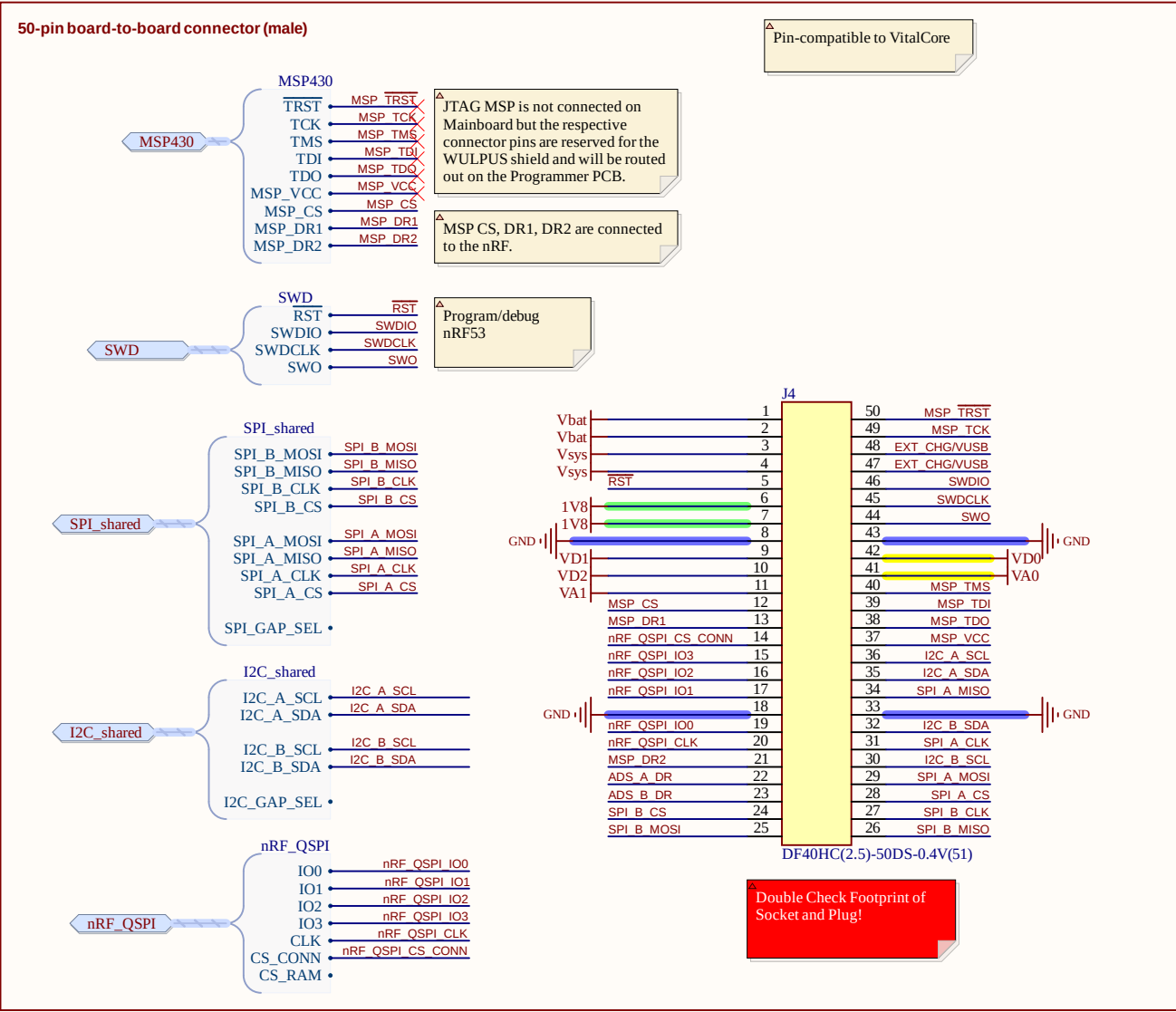
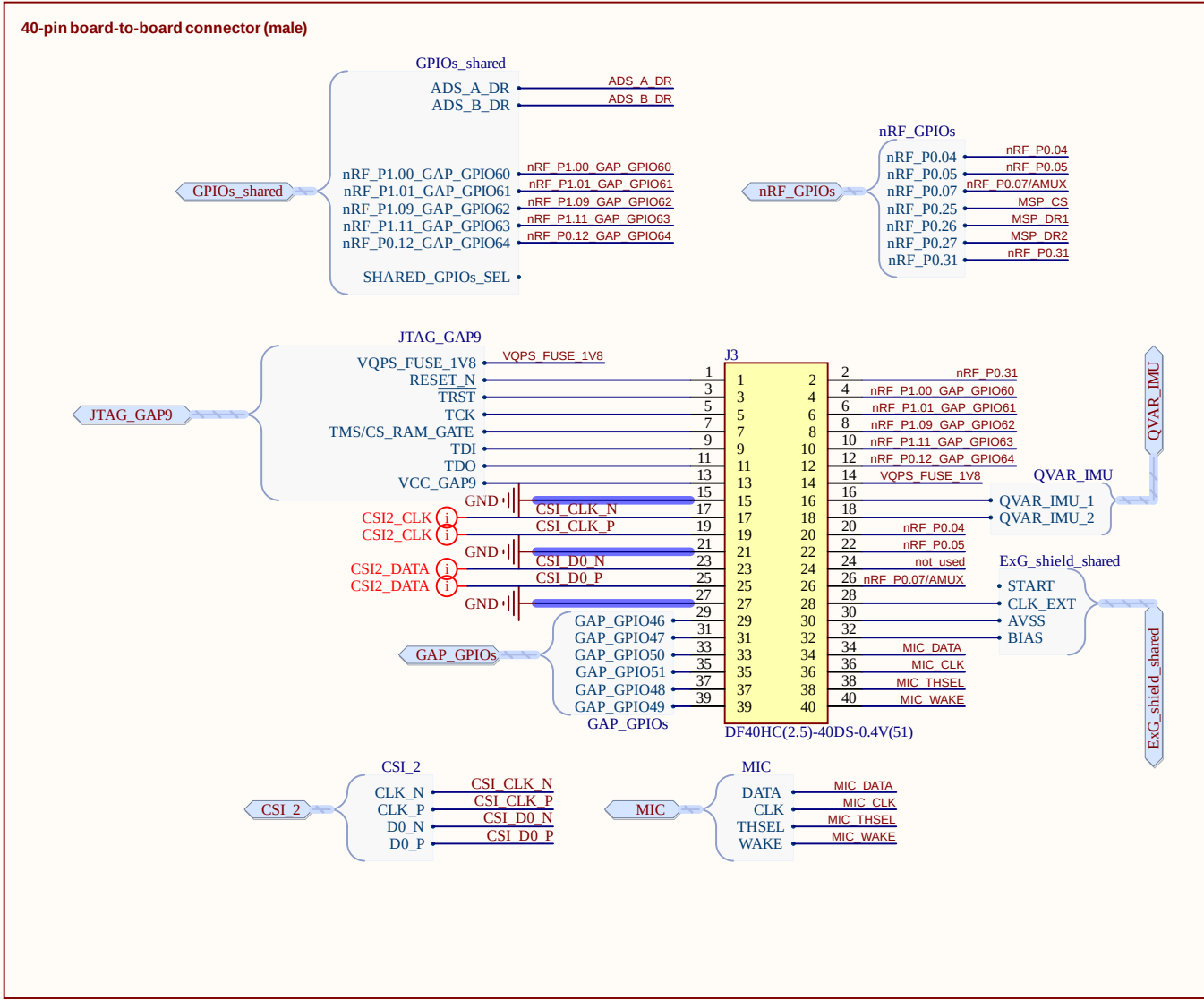
VA1 = 1.5V when bipolar supply is used
VA1 = 3.0V when unipolar supply is used



Digital supply for the ADS can be switched on/off with a load switch.





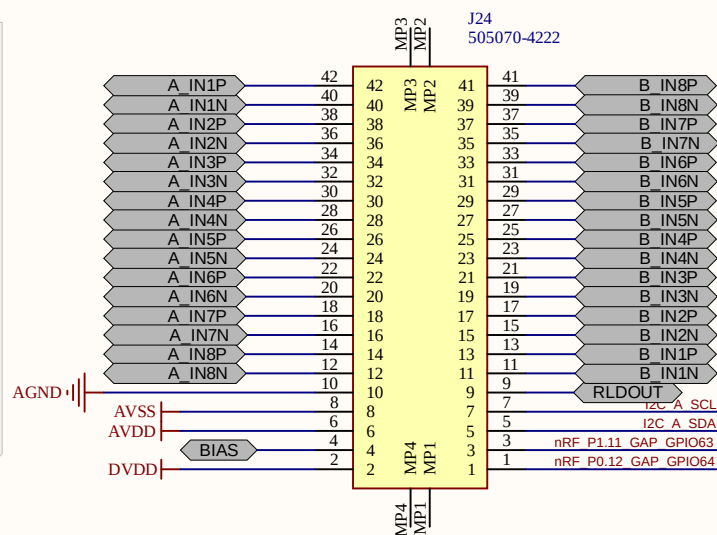


A

Board-to-board connector for fully differential config

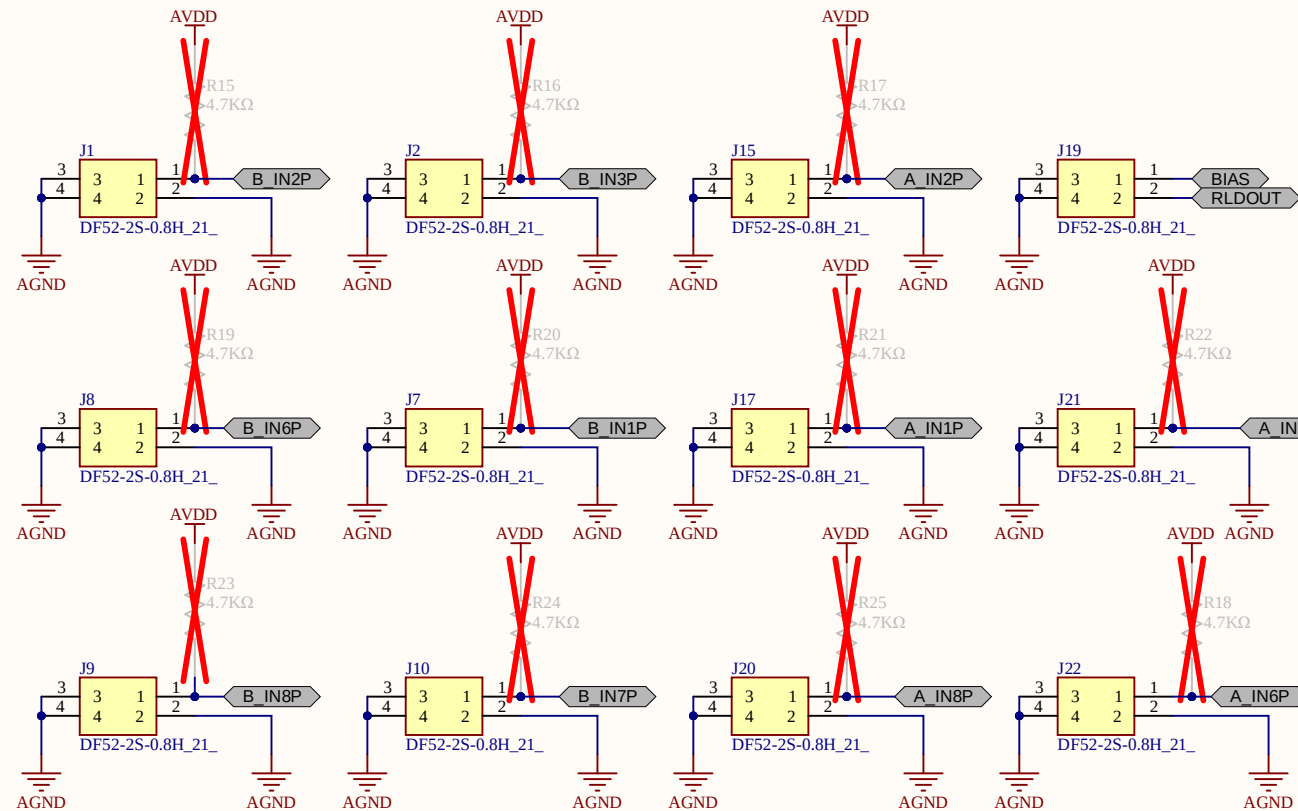
For fully differential configuration we need:

- Fully differential channels (16 max)
- AGND which is used to set the body bias in case of bipolar supply
- BIAS in case of unipolar supply and pullups
- AVDD and AVSS (in case of active electrodes)
- I2C for future use
- Two shared GPIOs for future use

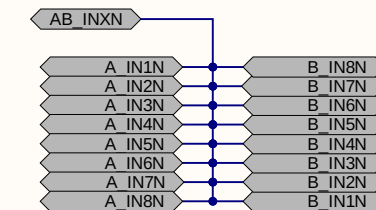


B

Break-off part



Connect together all neg ADS inputs for common reference. Only done on breakoff part of the PCB.

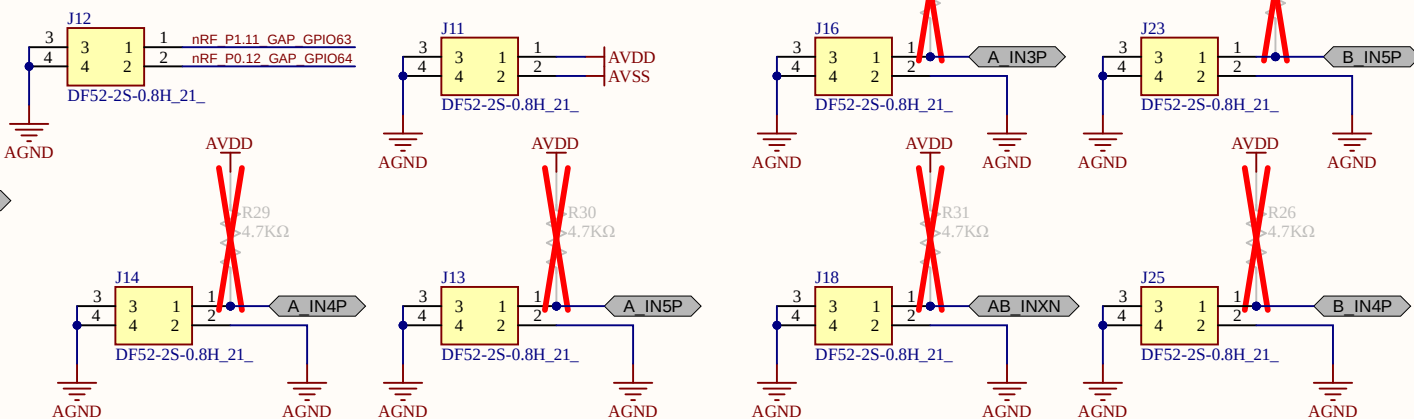


Breakoff part:

- Features a common REF for all negative inputs of the ADS and board-to-wire connectors.
- Can interface both passive and active electrodes (two-wire-config with pullups).
- Break this part away to use the fully differential configuration

For single ended configuration we need:

- Common reference (16 channels max)
- AGND which is used to set the body bias in case of bipolar supply
- BIAS in case of unipolar supply and pullups
- AVDD and AVSS (in case of active electrodes)
- Two shared GPIOs for future use



D



Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich

Project:

BioGAP ExG shield

Drawing number: *

Rev: v1.0

Format:

Laboratory: *

Sheet: 03_electrode_interface.SchDoc

Date: 23.09.2026 15:20:27

A3

Drawn by: *

Page * of *

File: C:\Users\ls-fre\Github\biogap_public\submodules\sensei-exg-shield\Hardware\ExG_shield_pcb\03_electrode_interface.SchDoc

A

B

C

D